

Godot

3 main reasons

To use Godot

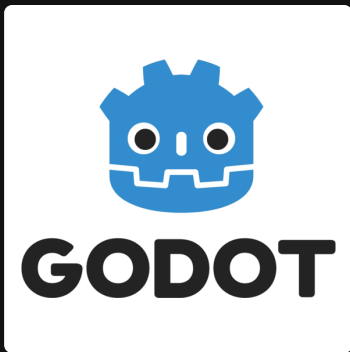
Knowing how tooling works

It's fairly rare to work on a project where you write **all** the code

Even in simpler project, we use libraries and plugins as a *shortcut* to get things done faster

In anything *larger*, we'll need to start working with **frameworks** and **engines** to get things done

The way you interact with any *framework or engine* is through is similar



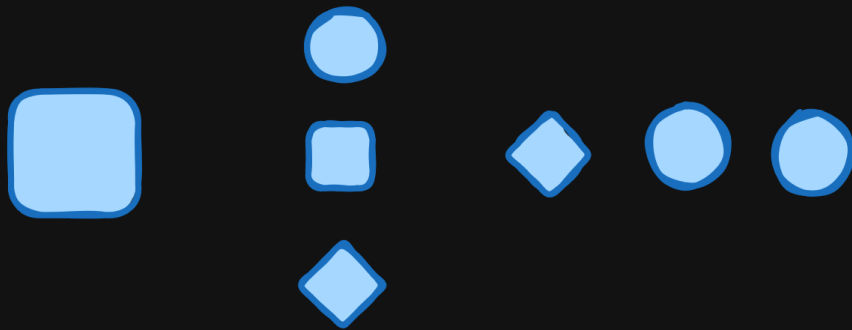
Godot is just a good example since it's all encompassing

Game development has more problem solving

In a lot of development, there are clearer "*right answers*" to problems

Even the *same problem* can have *multiple solutions*, and it's not always clear which one is best

This forces you to develop **raw problem solving skills**

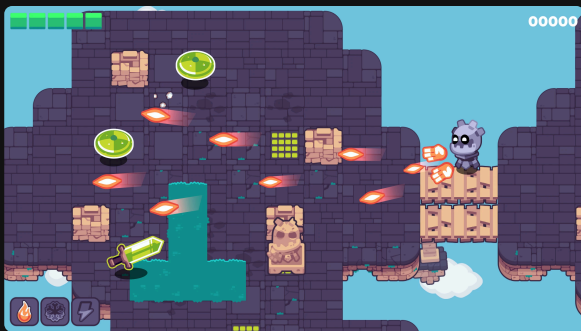


It let's you use **solutions**, gained from tutorials and courses, and the **tools** those solutions use

To solve *unique* problems using the same tools

Game dev is visual and interactive

This is a **huge** motivator for learning and improving



And also a great way to obtain an **instinct** for how things work

Programming can tend to be *abstract*, especially on the algorithms side

Game development allows you to *feel* and *see* the results of your work,

Which is a powerful feedback loop for learning and improvement

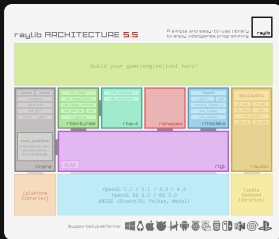
Why Godot

I don't technically have to sell you godot, but I will

It's easy to install, run, and get started with

Godot is a single executable, with *no installation process*, and no dependencies

It is **batteries-included**, meaning it has a lot of niceties built in, like a *code editor*, *file manager*, *game viewer*



Compared to *smaller engines*, like `raylib` and `love2d`,

Which require you to set up your own code editor, file manager, and game viewer

And Compared to *larger engines*, like `unreal` and `unity`,

Which have a much more complex installation process

It's free as in freedom and runs on anything

Godot is **free and open source software**, licensed under the MIT license



This means you can do whatever you want with it, and even contribute to the engine itself

As a student, this is important mainly:

1. You can use it for free, without worrying about *licensing or costs*
2. When you graduate, the *skills* you learned won't be behind a paywall

It also has really low system requirements, and can run on almost all platforms

It's pretty popular

It's been proven to work for a lot of different types of games, and has a large and active community



2025 showcase

Godot

Installation

Go to the Godot download page

Godot is a bit *unique* in the sense that it doesn't have an installation process, it's just a *single executable*

Unzip the downloaded file, and *run* the executable

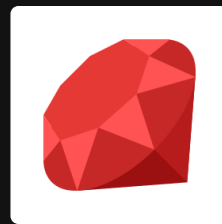
Godot as an Engine

Godot is a *game engine*, which means it provides a lot of *tools and features* to help you make *games*

And it runs using the `GDscript` programming language, which is a *Python-like* language designed specifically for Godot

A side note on programming languages

As a programmer, you **will** have to learn *multiple languages*



Most programmers are expected to know *at least a few languages*, be able to *learn new ones*, and be **good** at one or two

This is not as **difficult as it sounds**, because most programming languages share a lot of *common concepts* and *structures*

A mastery of basic programming concepts, like *variables*, *functions*, *classes*, *data structures*, and *algorithms* will allow you to pick up new languages quickly

A good resource is learn x in y minutes, which provides quick overviews of many programming languages

GDscript

All that to say, you will be learning `GDscript` to use Godot

It's *python-like*, emphasizes readability, and is designed to be easy to learn for beginners

But it's still a whole new language, with its own syntax, semantics, and quirks

We'll be using `Learn GDscript` by `GDQuest`, which is a great resource for learning the language and how to use it with Godot

Learn GDScript from Zero

Activity

In Learn gdscript, complete the first 9 lessons

Make a screenshot of your lesson progress

Submit it on neo